AI Adoption as a New Frontier of Regional Inequality in Korea

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ABSTRACT

Generative AI use in Korea is far more concentrated in the Seoul Metropolitan Area (SMA) than population, output or wages. Combining the Anthropic Economic Index (Claude.ai, April–May 2026) with the Regional Employment Survey (second half of 2025), this paper shows that the SMA accounts for 83% of Korean Claude usage but only 51% of employment. Work use per worker is 1.68 times the national rate in the SMA and 0.24–0.39 in the three other macro regions.

Occupational composition explains less than half of this gap. Within the same occupations, SMA usage is 2.4–3.1 times that of other regions, an order of magnitude more than the within-occupation gaps in wages or university degree shares (3–6%). The gap is as large among students, so it is likely to carry into the next cohort of workers. The exception is software, IT and data tasks, where usage is proportional to the relevant workforce in every region. This points to broader diffusion, rather than location as such, as the main policy lever.

1. Regional inequality in Korea

The SMA (Seoul, Incheon, Gyeonggi) now produces more than half of Korea's output on 12% of its land. Its share of nominal GRDP rose from 48.5% in 2000 to 52.8% in 2024, passing 50% in 2015. Its population share followed with a lag, from 46.3% to 51.0%, passing half in 2019 (Figure 1).

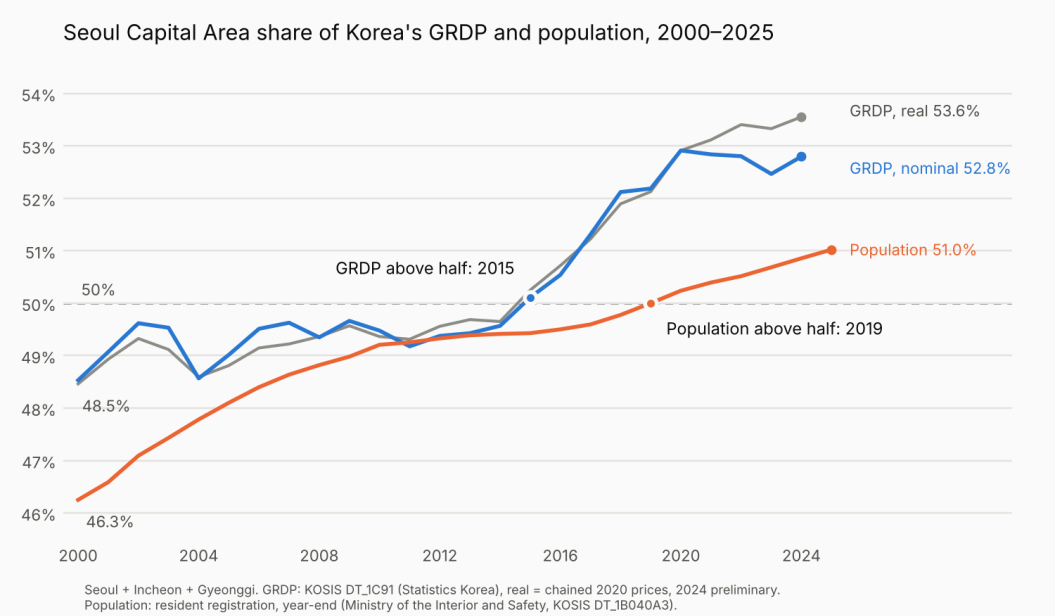


Figure 1. SMA share of GRDP (nominal and chained real) and of the resident population. Sources: Statistics Korea, Regional Income (KOSIS DT_1C91); Ministry of the Interior and Safety, resident registration (KOSIS DT_1B040A3).

Concentration accelerated after 2015 because the rest of the country slowed down, not because the SMA sped up. Real growth outside the SMA fell from 4.0% a year in 2000–2014 to 1.7% in 2014–2024, against 4.4% to 3.3% in the SMA. The SMA's share of national real growth rose from 51% to 67%. Had the rest of Korea kept its pre-2015 growth rate, the SMA's share would have fallen. The slowdown is concentrated in the industrial south-east and Jeolla, where shipbuilding, steel, chemicals and automobiles lost momentum, while Gyeonggi's semiconductor cluster kept growing (KDI 2026; Bank of Korea 2024).

Successive governments have tried to reverse this through balanced-development policy: the relocation of ministries to Sejong, the Innovation City programme that moved public agencies out of the capital, and industrial complexes in lagging regions. KDI estimates that, had only productivity changed between 2005 and 2019, the SMA's population share would have risen to about 62%. It stayed near 50% because amenities and the cost of absorbing population improved outside the capital region, partly through balanced-development investment in infrastructure and living conditions. These gains did not close the productivity gap that drives concentration.

2. AI as a new frontier of regional inequality

AI use is concentrated in the SMA well beyond its share of people or jobs. Korea as a whole is a heavy Claude user, ranking 14th of 121 countries in usage per working-age person. Within Korea, the SMA accounts for 83% of usage against 53% of the working-age population and 51% of employment.

The gap is starkest for work use. Measured as a region's share of Korean work-related usage divided by its share of employment, the SMA uses Claude at 1.68 times the national rate, against 0.39 in Chungcheong, 0.29 in Gyeongsang and 0.24 in Jeolla (Figure 2). Outside the SMA, only the administrative capital Sejong (1.08) and the research hub Daejeon (0.75) come close; the industrial provinces Gyeongnam, Gyeongbuk and Jeonnam sit at 0.13–0.17.

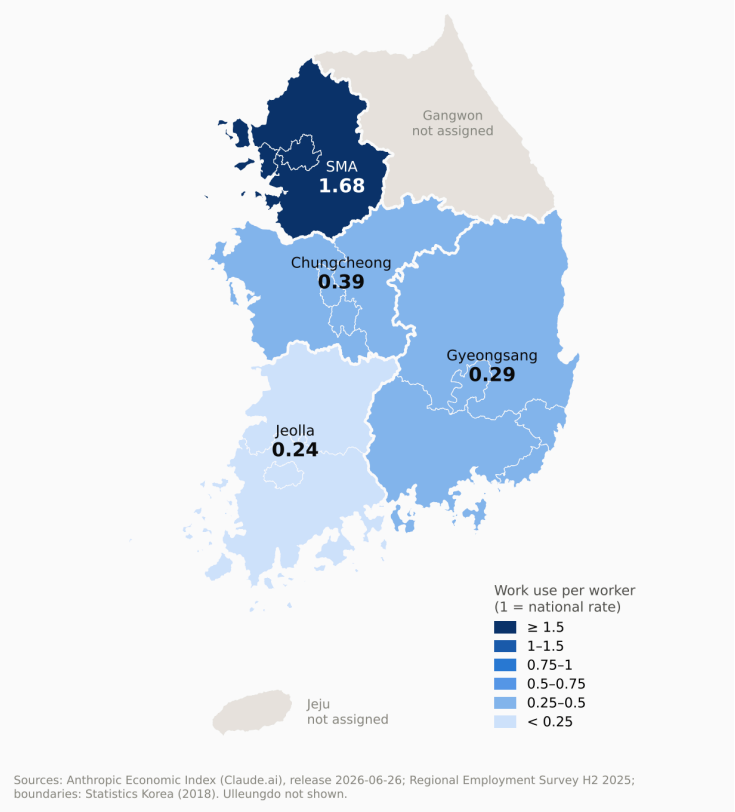


Figure 2. Work use of Claude per worker by macro region: share of Korean work-related Claude usage divided by share of employment (by workplace), April–May 2026; 1 = national rate. Gangwon and Jeju (grey) are not assigned to a macro region because their survey samples are too thin for within-occupation comparisons (see Data and methods). Sources: Anthropic Economic Index; Regional Employment Survey H2 2025; boundaries: Statistics Korea (2018).

The same divide appears in survey data on any generative AI, though less extreme: 52–53% of internet users in Seoul, Gyeonggi and Incheon used generative AI in 2025, against 25–34% in Jeonbuk, Gyeongbuk, Busan and Gyeongnam (MSIT 2025 Internet Usage Survey).

3. Not a continuation of existing differences

The SMA's lead is not simply a matter of where AI-intensive jobs are located. The SMA does hold more of the occupations whose tasks Claude is used for most, such as software, media, business and management. If every region used Claude at the national rate within each occupation, the SMA would account for 64% of Korean usage. It actually accounts for 83%, a remaining factor of 1.30.

After this adjustment the other macro regions stay far behind, at 0.42 (Jeolla) to 0.54 (Chungcheong) of the national rate (Figure 3). Occupational composition accounts for about 47% of the SMA's excess over its employment share; the other 53% is higher usage within the same occupations. Across all regions, the adjustment reduces the share of usage that would need to move to match employment from 32% to 18%.

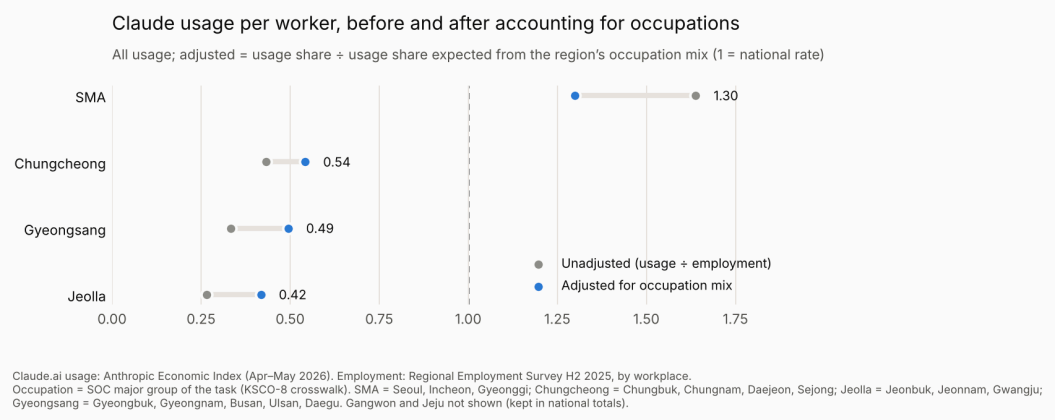


Figure 3. Grey: usage share ÷ employment share. Blue: usage share ÷ the share expected from the region's occupation mix at national usage rates (1 = national rate). Occupations are SOC major groups, mapped from KSCO-8 two-digit codes.

The within-occupation gap is widest in everyday office work. In Educational Instruction, Office and Administrative Support and Sales, SMA usage is 1.5–1.6 times the national rate, against 0.26–0.46 elsewhere. Other worker characteristics do not close it either. Men and women in work are spread across regions almost identically, and within occupations workers outside the SMA are at most 1.3 years older on average.

4. Why this could amplify regional inequality

The AI gap is an order of magnitude larger than the regional gaps it might be expected to mirror. Comparing workers in the same occupation, hourly wages outside the SMA are 94–97% of the SMA level and university degree shares 94–97%. Claude usage is 32–42% (Figure 4). Adding age controls leaves the wage and education ratios essentially unchanged.

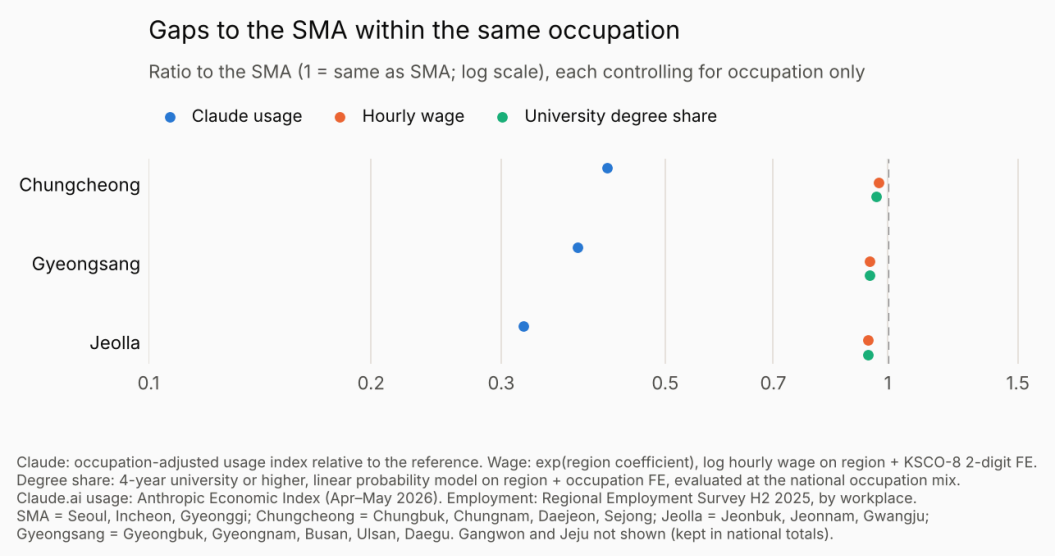


Figure 4. Ratio to the SMA on a log scale, each controlling for occupation only. Claude: occupation-adjusted usage index. Wage: log hourly wage on region and KSCO-8 two-digit occupation fixed effects. Degree share: four-year university or higher, linear probability model with the same controls.

If AI raises the productivity of the people who use it, today's adoption gap becomes tomorrow's productivity and wage gap. That would work on precisely the margin that has driven concentration since 2015 and that balanced-development policy has not moved. The gap is not confined to the current workforce.

Coursework use per student, measured against enrollment in high schools and higher education, is 1.56 in the SMA against 0.35–0.52 elsewhere, and almost identical when only higher-education students are counted (Figure 5). Students outside the SMA are entering the labour market with less experience of these tools.

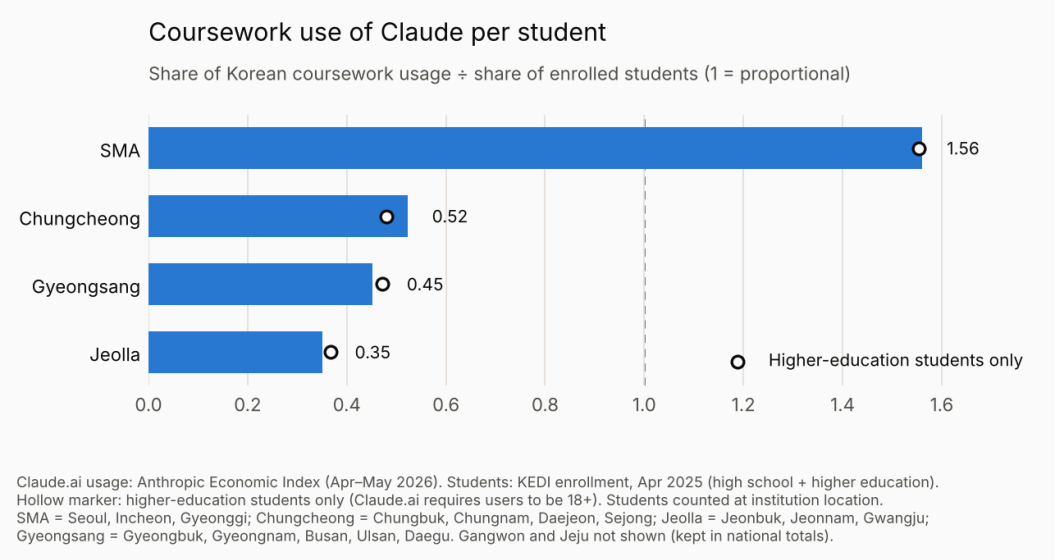


Figure 5. Share of Korean coursework usage ÷ share of enrolled students (KEDI, April 2025). Hollow markers: higher-education students only. Students counted at institution location.

5. Broader diffusion as a lever

In the most technology-intensive occupations, the SMA does not outperform. Usage of Computer and Mathematical tasks (software, IT, data and document handling) is proportional to the relevant workforce in every macro region: 0.98 in the SMA, 0.95 in Chungcheong, 1.09 in Jeolla and 1.15 in Gyeongsang (Figure 6). In every other large occupation group the SMA leads: by a factor of 1.3–2.1 in Arts and Media, and 2.4–6.1 in education, office, sales and business occupations.

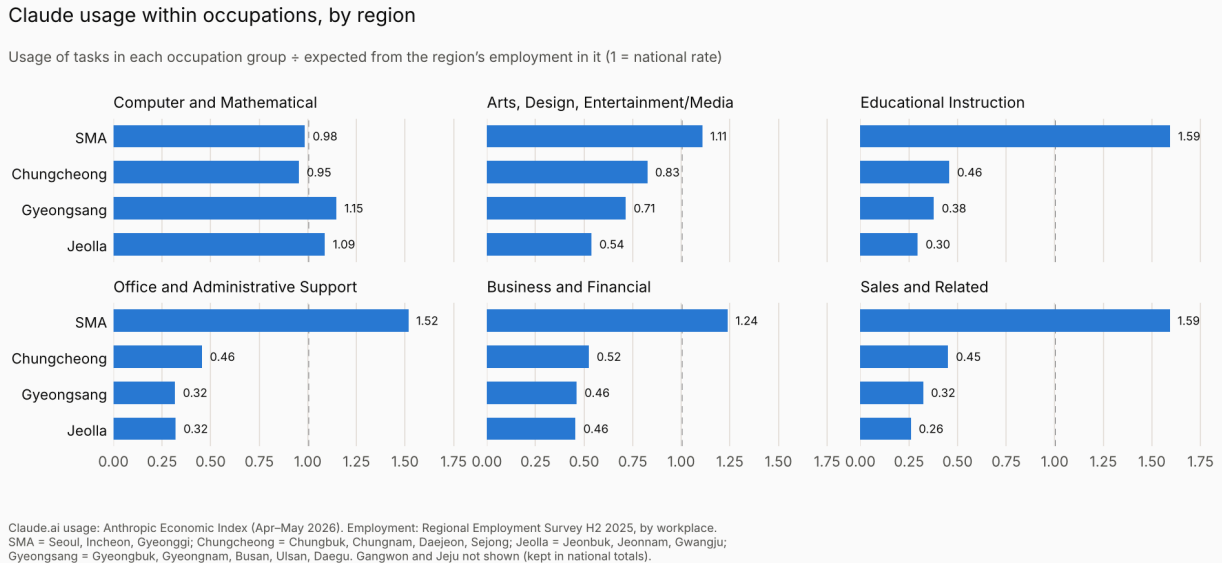


Figure 6. Usage of tasks in each occupation group ÷ usage expected from the region's employment in it (1 = national rate).

This pattern is consistent with technology-native workers adopting AI wherever they are, while adoption in other occupations is spreading outward from the capital. It also suggests the gap is one of adoption rather than depth. Users outside the SMA work with Claude in much the same way: average AI autonomy is 2.62–2.64 on a five-point scale in all four macro regions, the automation share 43–45% and the estimated speed-up on tasks 6.8–7.1 times.

The policy implication is that regional AI policy should focus on breadth of adoption: access, awareness and firm-level uptake in office, education, sales and business occupations outside the capital. Unlike the productivity gaps that place-based policy has struggled to close, an adoption gap among workers already doing the same jobs is something targeted diffusion can address directly.

6. Conclusion

Korea's regional divide has a new dimension. Generative AI use is concentrated in the SMA far beyond its shares of population, employment or output, and the concentration survives accounting for occupations, wages, education, age and gender. Within the same jobs, workers outside the capital use Claude at between a third and two fifths of the SMA rate, while earning about 95% of SMA wages and holding degrees at similar rates.

Because AI is a general-purpose productivity technology, a gap of this size risks widening the productivity differences that have driven concentration since 2015, and it is already visible among students. The even distribution of software and IT usage, and the similar way in which users everywhere work with Claude, suggest that the gap reflects slower diffusion rather than fixed regional disadvantage. Regional policy that targets breadth of adoption in non-technical occupations outside the capital may therefore achieve more than another round of infrastructure investment. Longer time series, firm-level adoption data and more precise sub-national location data would allow these patterns to be tested causally.

Data and methods

- **AI usage:** Anthropic Economic Index, release of 26 June 2026, Claude.ai (Free, Pro and Max) conversations, averaged over April and May 2026. Province-level data give each province's share of Korean usage, its use-case shares (work, personal, coursework), usage measures such as autonomy, and its occupation mix as SOC major groups. Conversations are located by IP address.
- **Employment, wages, education, age:** Statistics Korea Regional Employment Survey, second half of 2025 (October), about 246,000 employed respondents, weighted. Region is the province of the workplace. Hourly wage = three-month average monthly wage ÷ (weekly hours × 4.345), trimmed at the 0.5th and 99.5th percentiles.
- **Occupations:** KSCO-8 two-digit codes mapped to the 22 SOC major groups with a judgment-based crosswalk; codes spanning several groups are split with fixed weights.
- **Occupation adjustment:** the benchmark allocates national usage in each SOC group across regions by their employment in that group; the index is actual usage ÷ benchmark (1 = national rate).
- **Students:** KEDI enrollment counts, 1 April 2025, by institution location (high school, undergraduate, master's, doctoral).
- **GRDP and population:** KOSIS tables DT_1C91 (Regional Income) and DT_1B040A3 (resident registration, year-end).
- **Regions:** SMA = Seoul, Incheon, Gyeonggi; Chungcheong = Chungbuk, Chungnam, Daejeon, Sejong; Jeolla = Jeonbuk, Jeonnam, Gwangju; Gyeongsang = Gyeongbuk, Gyeongnam, Busan, Ulsan, Daegu. Gangwon and Jeju are excluded from the macro comparisons but kept in national totals. Each is a single province with a thin survey sample for within-occupation comparisons: Jeju has 1,973 wage earners in the Regional Employment Survey, with 28 of its 52 occupation cells below 30 observations, and in Gangwon 16 of 56 occupation cells fall below 30, against at most 7 in each macro region.

Limitations. IP-based location is imprecise below the country level and may assign some traffic to Seoul. The Index's occupations describe the task in a conversation, not the user's job. Two months of data allow only cross-sectional comparisons, and small cells are suppressed in the published data.

Sources

- Anthropic Economic Index report: Cadences (June 2026)
- Anthropic Economic Index report: Economic Primitives (January 2026)
- KDI Focus: why capital-region concentration continues (January 2026)
- Bank of Korea regional economic report on capital-region growth contribution (March 2024)
- Bank of Korea Issue Note 2024-15
- MSIT 2025 Internet Usage Survey, regional results